MATH 201 Your name: ______________________________

Learning target DF4, version 1

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(x) = \dfrac{\ln(x)}{2x^2 + 6x - 3}$

2. $f(x) = \left(x^2 - 2x - 4\right)\cos(x)$

3. $h(t) = -\dfrac{3t^2 + 2t + 3}{t^3}$

MATH 201 Your name: ____________________

Learning target DF4, version 2

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $h(w) = -\frac{\cos(w)}{6w^2+5w+4}$

2. $g(w) = \frac{3w^2+5w-1}{w^8}$

3. $f(w) = -\left(4w^2+6w+1\right)\ln(w)$

MATH 201 Your name: ______________________________

Learning target DF4, version 3

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $g(w) = -\dfrac{2w^2 + 6w + 1}{w^{1/5}}$

2. $f(w) = -\dfrac{\sin(w)}{3w^2 - 3w + 4}$

3. $h(w) = -\left(2w^2 + 5w - 5\right)e^w$

MATH 201 Your name: ____________________

Learning target DF4, version 4

Compute derivatives using the Product and Quotient Rules.

Demonstrate and explain how to find the derivative of the following functions. Be sure to write down which derivative rules (product, quotient, sum and difference, etc.) you are using.

1. $f(x) = -\left(3x^2 - 6x + 2\right)\cos(x)$

2. $h(y) = \dfrac{3y^2 - 3y - 2}{\sin(y)}$

3. $g(w) = \dfrac{2w^2 - w + 4}{w^{1/6}}$